

Beautified Is Not Forged: Supplementary Material

S1. FIGURES AND DETAILS MOVED FROM THE MAIN PAPER

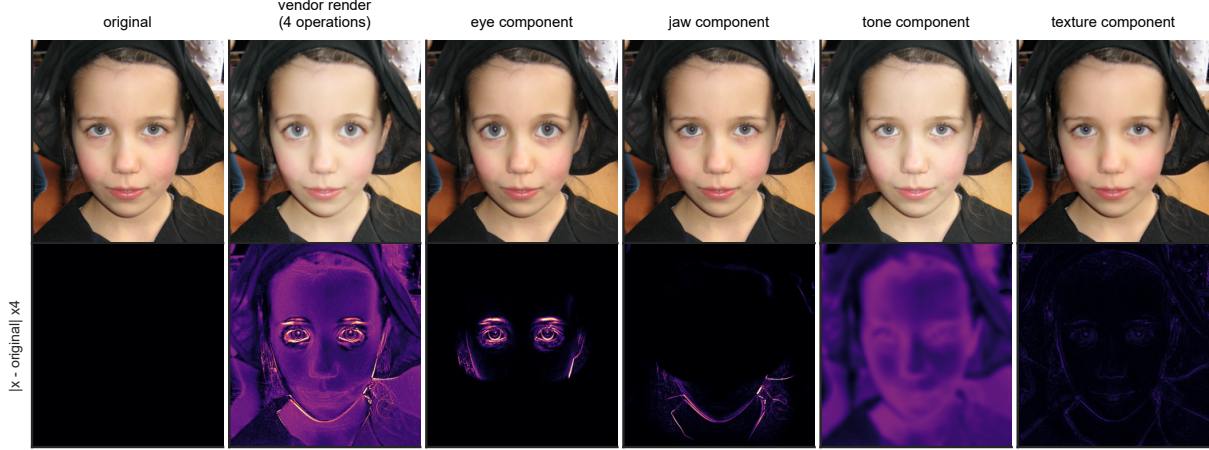


Fig. S1. Splitting a four-operation Tencent render into single-operation components in the original’s geometry. Bottom: absolute difference to the original, $\times 4$. Eye and contour components are the render’s optical flow restricted to the eye regions and to the rest of the face; tone and texture components are the low- and high-pass parts of the photometric residual. The four parts recompose the render.

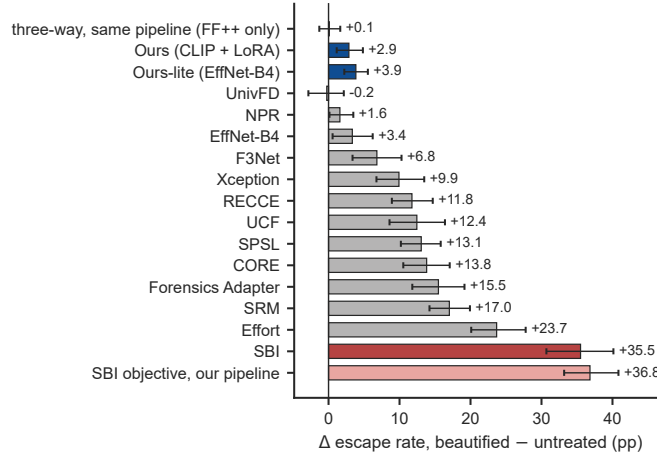


Fig. S2. Celeb-DF-B: change in the share of deepfake frames called real when the same videos are beautified, with 95% video-cluster intervals. Binary detectors at 5% FPR on untreated genuine frames; three-way models at their native rule.

Measurement pitfalls. (1) An earlier Celeb-DF-v2 set of ours (389 frames from 307 videos, filtered by face-detector confidence) placed every published detector 3–10 AUC points above its published frame-level number; on the standard list our scores are close to the published ones (Xception 0.718 vs. 0.737, EfficientNet-B4 0.750 vs. 0.749, UCF 0.744 vs. 0.753). (2) Test crops must match the training extractor (landmark hull, 35% margin, aspect kept); a square $1.3\times$ crop cost our model 1.8 points. (3) Celeb-DF-B’s treated videos are re-framed to portrait with filter-tinted bars, and the landmark detector then misses 57% of treated frames against 3% of untreated ones; trimming the bars restores 97% coverage.

SBI checkpoint. Our reproduction of the SBI recipe reached 0.735 on Celeb-DF-v2 against 0.817 for the released weights, so every SBI comparison uses the released checkpoint.

Paired controls. Without its unedited control a filter rate is a base rate: a model trained on mixed corpora called 96.8% of RetouchingFFHQ retouched faces *filter*, and 93.9% of their untouched originals.

Edge model. A RepViT-M0.9 version trained with the vendor renders keeps the commercial-retouching behaviour (balanced 65.1%, retouched→fake 1.1%) at 4.7M parameters, exports to TFLite (18MB, fp32) with 200/200 decisions identical to

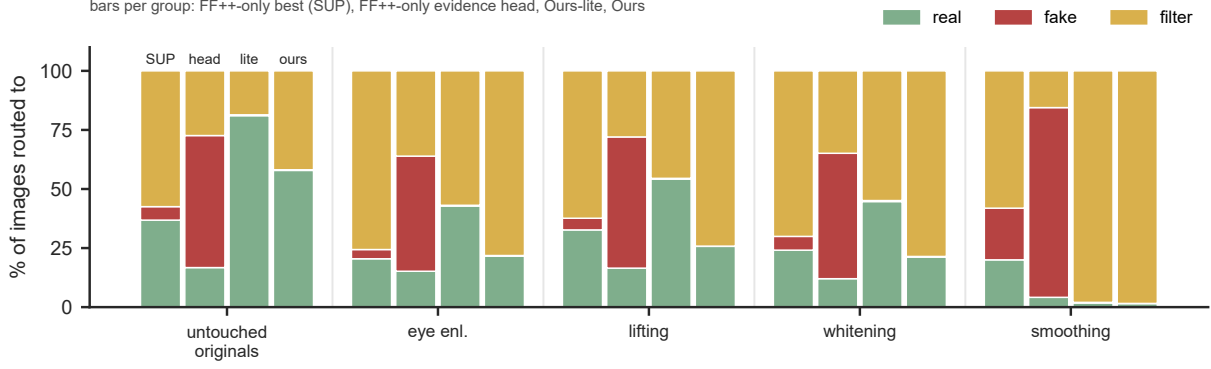


Fig. S3. Where the unseen service’s retouched faces go (levels pooled): share routed to real, fake and filter for the best FF++-only three-way model, the FF++-only evidence-head model, Ours-lite and Ours.

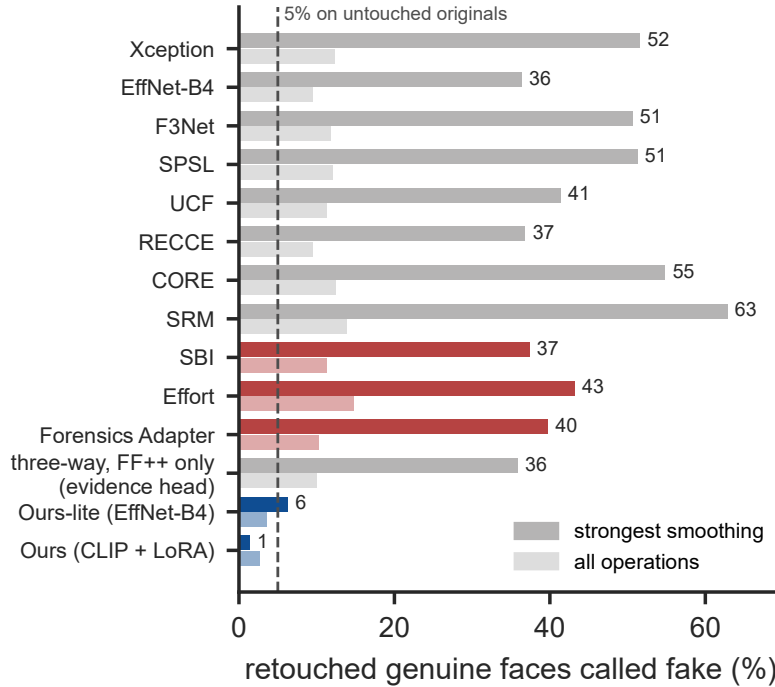


Fig. S4. Retouched genuine faces from the unseen service called *fake*, each detector at the threshold that calls 5% of the untouched originals fake (chosen on the test originals, which favours the binary detectors). Dark: strongest smoothing; light: all operations and levels. Red: SBI and the two CLIP-based detectors.

PyTorch, and runs at 9.7 ms per image on one CPU thread. Trained with the single-operation components it calls 55.7% of untouched originals *filter*.

Edit-magnitude head. A head regressing the measured quantities q ranks the measured edit at Spearman $\rho \leq 0.47$ for every operation on the unseen service and barely better on the training services; it is not used.

S2. TRAINING CURVES AND ACCURACY PER PARTITION

Both checkpoints are chosen on the validation data; no test result enters the choice. Ours reaches its best validation score at epoch 4 of 8 and Ours-lite at its last epoch, 12. On FF++ the genuine class carries most of the gap between training and test: Ours recalls 94.2% of genuine training frames and 78.0% of genuine test frames, and the confusion matrices show these frames moving to both other classes.

S3. ABLATION OVER THE TRAINING SIGNAL

Why single-operation components. A model trained on the renders alone sends single operations other than smoothing to *filter* only 10–46% of the time. On held-out Tencent and Megvii bases a whitening component with a lightness change near 20 is then called real 90% of the time, because in a four-operation render the texture change alone identifies the class.

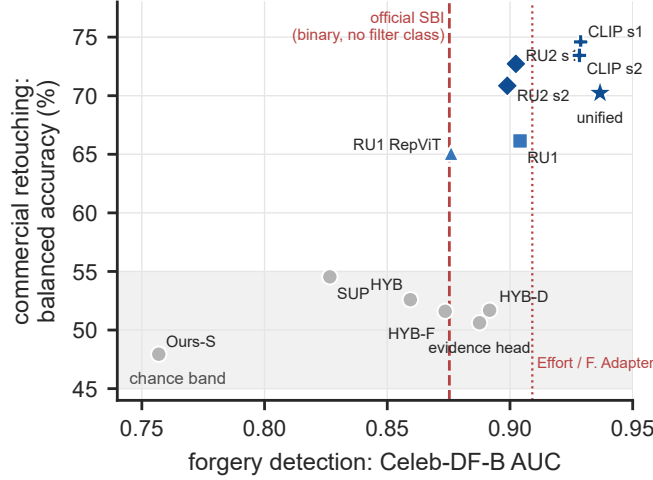


Fig. S5. Untreated Celeb-DF-B AUC against balanced accuracy on the unseen service, for every three-way variant. Binary detectors have no filter class and appear as vertical lines.

TABLE S1
EXPLANATIONS ON HELD-OUT PART-LEVEL FORGERIES, ALL VARIANTS (COLUMNS AS IN THE MAIN PAPER).

Map	donor transplant				SD-1.5 inpaint				SDXL inpaint (unseen editor)			
	IoU ₇	flip	ceil.	compl.	IoU ₇	flip	ceil.	compl.	IoU ₇	flip	ceil.	compl.
centre prior / random region (last col.)	0.310	38.6	–	17.0	0.249	40.9	–	17.9	0.255	40.2	–	17.7
no head, Grad-CAM	0.367	50.9	85.8	15.2	0.259	45.4	85.2	25.0	0.269	48.0	85.9	23.6
no head, Grad-CAM++	0.414	51.7	85.8	7.2	0.327	47.8	85.2	16.7	0.330	48.8	85.9	19.1
no head, Score-CAM	0.408	56.5	85.8	5.9	0.334	52.4	85.2	15.4	0.341	52.4	85.9	17.1
head, part edits, FF++ only	0.723	84.0	84.7	0.4	0.752	82.6	83.2	0.7	0.746	82.9	83.8	0.7
+ symmetric degradation	0.723	86.3	87.1	0.4	0.719	82.5	83.7	2.0	0.728	83.7	84.4	1.5
+ restoration-consistency loss	0.759	89.4	90.0	0.4	0.614	75.2	79.8	4.4	0.684	82.5	84.6	2.6
+ consistency + symmetric degradation	0.722	88.4	89.3	0.7	0.723	85.2	86.1	2.4	0.732	86.1	86.9	1.8
head, vendor data, no part edits (EffNet-B4)	0.418	61.4	82.4	12.0	0.280	39.5	63.5	23.4	0.315	46.2	72.3	22.5
head, vendor data, no part edits (CLIP)	0.479	78.8	89.4	2.5	0.376	53.9	69.1	14.4	0.445	68.9	78.3	8.4
Ours-lite (EffNet-B4)	0.804	87.8	87.8	0.1	0.820	87.4	87.6	0.4	0.812	86.9	87.2	0.4
Ours (CLIP ViT-L/14 + LoRA)	0.832	91.7	91.9	0.1	0.845	91.5	91.6	0.2	0.827	90.9	91.2	0.5

S4. EVERY VENDOR-TRAINED VARIANT ON THE UNSEEN VENDOR

S5. ESCAPE PER BEAUTIFICATION PRESET

The grain filter (*hawaii_grain*) is the most damaging for binary detectors (+9 to +56 pp) and lowers our escape rate; the warm colour grading (*california_dreamin*) is our failure case (HYB +10.1 / +11.9 pp for two seeds, RU1 +8.5 pp), comparable to SRM and CORE.

S6. FEATURE SPACE

S7. WHAT THE DETECTOR LOOKS AT: REGION RESTORATION

Grad-CAM peaks at the face centre for every class, and occlusion with a constant fill injects a blending-like seam that itself pushes the model toward *fake*. Restoring a region to its pixel-aligned original is a cleaner test. Eye enlargement is carried by the eyes (0.47 vs 0.18 for a random patch of equal area); face slimming by the pixels the warp displaces around the jaw. Skin smoothing is carried mostly by the *background* (0.74) rather than the skin (0.08): our scripted smoothing acts on the whole frame, and the detector uses the smoothed background. For forgeries no single region carries the decision (every region ≤ 0.12), while restoring the whole frame removes it (0.70–0.82).

S8. UNSEEN PHOTOMETRIC FILTERS

On 11 photometric filters never seen in training, applied to FF++ test frames, HYB routes 51.8% of frames to *filter* against 7.4% for the unedited frames (paired excess 44.7–45.5 pp) and calls 9.5% of them *fake*; the RepViT model’s excess is ≈ 52 pp with 4.6% fake calls. RU2 keeps a paired excess of 31–34 pp.

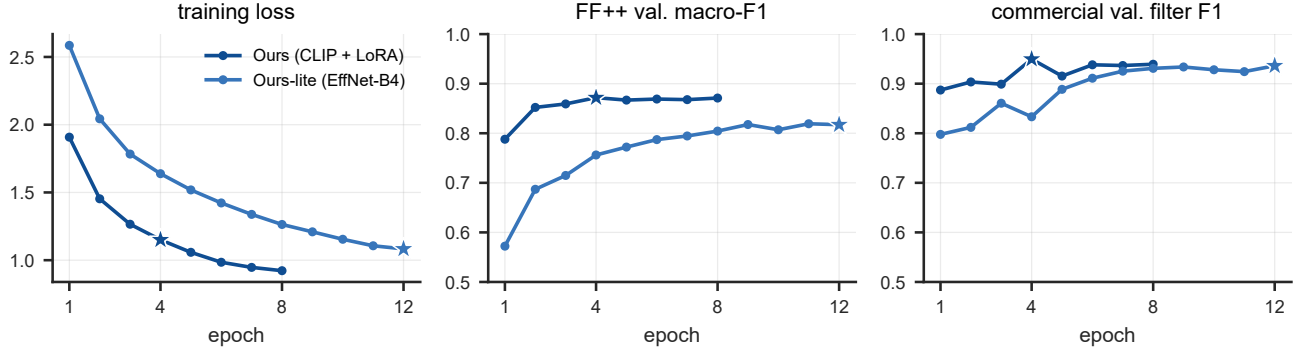


Fig. S6. Training loss and the two validation scores per epoch for Ours and Ours-lite. Star: the selected checkpoint, the epoch with the highest mean of FF++ validation macro-F1 and the filter F1 on held-out Tencent and Megvii bases. The commercial validation F1 here counts every validation component, including those under the noise floor that training drops, so it differs from the cleaned rows of the partition table in the main paper.

training sample (n = 6,000)				validation (n = 1,589)				test (n = 7,936)				
Ours (CLIP + LoRA) true class	real	94.2	0.9	4.9	real	80.9	7.5	11.5	real	78.0	8.8	13.2
	fake	7.3	89.6	3.1	fake	7.4	90.7	1.9	fake	6.4	88.9	4.8
	filter	8.1	0.7	91.3	filter	8.1	1.9	90.0	filter	6.8	4.0	89.2
		real	fake	filter		real	fake	filter		real	fake	filter
(n = 6,000)				(n = 1,589)				(n = 7,936)				
Ours-lite (EffNet-B4) true class	real	96.9	1.3	1.8	real	78.5	10.6	10.9	real	74.2	11.6	14.1
	fake	5.8	94.0	0.3	fake	6.2	92.6	1.1	fake	4.1	93.3	2.6
	filter	13.8	2.9	83.3	filter	14.9	10.0	75.1	filter	11.9	11.7	76.4
		real	fake	filter		real	fake	filter		real	fake	filter
		predicted				predicted				predicted		

Fig. S7. FF++ three-way confusion matrices, rows normalised to the true class (%): a random sample of the training frames (2,000 per class, fakes 500 per method), the validation split and the test split.

S9. EVIDENCE HEAD: DETECTION, PER-SOURCE RECALL, ROBUSTNESS, COST, OUTPUT

NeuralTextures, whose edits are confined to the mouth, is the weakest forgery method (73.9%); eye enlargement at our scripted strengths is recognised at 37%. Under blur or downscaling the models without symmetric degradation collapse on genuine frames (2% at $\sigma = 1.5$) while HYB-D keeps 66–83%. The head adds 0.01 M parameters and 7 ms on CPU.

S10. SYSTEM OUTPUT OF OURS

The facial part is named by the mean evidence inside each landmark region. On held-out part-level forgeries (every fourth item, detected ones) this names the edited part in 99.0%, 98.5% and 97.7% of cases for the donor, SD and SDXL editors. Naming it by the share of the top-12% evidence region that falls on each part, as in an earlier version, names it in only 8–12%, because the skin region is large; the map itself is unchanged. Operation names are reliable on the training services' held-out faces for whitening and smoothing (presence recall 0.79–0.99) and weaker for eye enlargement and face reshaping (0.50–0.80); on the unseen service they drop to 0.21 (eyes), 0.35 (reshaping), 0.57 (whitening) and 0.96 (smoothing). On 379 fixed test images the template produces 14 distinct sentences: one for *real*, four for *fake* and nine for *filter*. The edit-magnitude estimates are not shown because they did not pass validation (Sec. S1).

TABLE S2

ABLATION OVER THE TRAINING SIGNAL. FF++ IN-DOMAIN RECALL PER CLASS (%), FA/MISS ON THE FF++ BEAUTIFICATION SETS, AND B1: NUMBER OF BINARY DETECTORS (/11 PUBLISHED, /12–13 WITH OUR SBI REPRODUCTIONS) THAT DOMINATE THE MODEL.

Variant	Fakes seen	CDFv2	DFD	real	fake	filter	FA / MISS	B1
Ours-S (RepViT)	observed	0.701	0.888	85.6	91.2	86.0	2.7 / 1.2	0/11
Ours-SB (self-blend only)	self-blend	0.628	0.879	95.8	32.2	36.2	7.4 / 22.3	6/12
Ours-SUP (observed only)	observed	0.784	0.912	80.7	95.0	88.4	4.9 / 0.5	0/13
Ours HYB, seed 1	both	0.757	0.917	85.9	94.4	84.7	5.8 / 0.6	0/12
Ours HYB, seed 2	both	0.770	0.913	87.6	92.8	84.3	4.9 / 0.7	0/12
Ours HYB-F (faithful blend)	both	0.779	0.900	82.6	93.8	84.7	6.1 / 0.5	0/13
Ours HYB-D (+ symmetric degradation)	both	0.807	0.882	81.6	91.8	83.7	5.2 / 0.8	0/13
Ours HYB-PE (+ part-level edits)	both+parts	0.810	0.888	80.7	90.2	76.8	8.9 / 1.6	0/13
Ours HYB-PE + evidence head	both+parts	0.819	0.907	79.0	90.0	84.0	7.0 / 2.1	0/13
Ours-RU1 (+ vendor composites)	both+vendor	0.826	0.929	78.0	93.3	80.7	8.7 / 0.4	0/15
Ours-RU2, seed 1	both+vendor	0.822	0.938	79.4	93.6	75.8	10.3 / 0.5	0/15
Ours-RU2, seed 2	both+vendor	0.817	0.934	80.4	92.9	76.1	9.9 / 0.5	0/15
Ours-RU3 (cleaned components)	both+vendor	0.832	0.938	79.8	93.7	74.9	10.4 / 0.6	0/15
Ours-RU3 + evidence head	both+vendor	0.824	0.930	79.9	94.0	76.4	10.2 / 0.3	0/15
Ours-RU3 + CLIP-L/14 LoRA, seed 1	both+vendor	0.878	0.957	76.7	94.6	87.9	5.5 / 0.2	0/15
Ours-RU3 + CLIP-L/14 LoRA, seed 2	both+vendor	0.883	0.953	87.2	92.2	86.9	4.1 / 0.5	0/15
Ours-RU4 + evidence head (EffNet-B4)	both+vendor+parts	0.848	0.915	74.2	93.3	76.4	11.7 / 0.5	0/15
Ours-RU4 + CLIP-L/14 LoRA (unified)	both+vendor+parts	0.888	0.948	78.0	88.9	89.2	4.0 / 0.6	0/15
SBI recipe, our pipeline (binary)	self-blend	0.620	0.852	–	–	–	17.5 / 28.7	–
SBI recipe, faithful blend (binary)	self-blend	0.735	0.886	–	–	–	19.3 / 20.2	–

TABLE S3

ALIBABA PAIRS, ALL VENDOR-TRAINED VARIANTS (SINGLE READS). COLUMNS AS IN THE MAIN PAPER.

Model	originals	retouched (pooled)		balanced	paired rank	paired rank at level 90				smoothing-90
	real/fake/filter	→filter	→fake			eye / white / lift / smooth	→fake			
<i>FF++ c23 training only (zero-shot)</i>										
three-way, RepViT-M0.9	0.8/2.5/96.7	92.6	6.4	47.9	48.4	73 / 55 / 52 / 15				17.7
three-way, observed fakes (SUP)	36.8/5.7/57.5	66.5	9.1	54.5	79.8	97 / 89 / 79 / 59				27.3
three-way, hybrid fakes (HYB)	33.8/16.7/49.5	54.6	22.1	52.6	69.2	92 / 83 / 65 / 44				50.2
HYB, faithful blend	28.2/22.1/49.6	52.8	28.0	51.6	65.7	92 / 77 / 62 / 35				58.4
HYB + symmetric degradation	31.7/26.3/42.0	45.3	33.6	51.7	64.3	89 / 76 / 63 / 35				63.1
three-way, part edits + evidence head	16.7/55.9/27.4	28.6	59.4	50.6	60.9	89 / 78 / 60 / 20				85.8
<i>Vendor-trained variants (all single Alibaba reads)</i>										
RU1 renders only (EffNet-B4, s1)	93.4/0.8/5.8	38.1	0.6	66.1	92.8	99 / 85 / 95 / 100				0.1
RU1 renders only (RepViT)	92.8/1.4/5.9	36.1	1.1	65.1	92.6	96 / 89 / 93 / 100				0.3
RU2 + components (EffNet-B4, s1)	76.8/0.5/22.8	68.2	0.3	72.7	94.2	99 / 90 / 94 / 100				0.4
RU2 + components (EffNet-B4, s2)	66.6/0.3/33.2	74.9	0.2	70.8	94.2	99 / 92 / 93 / 99				0.4
RU2 + components (RepViT)	43.4/0.9/55.7	83.7	0.5	64.0	95.1	99 / 91 / 96 / 99				1.1
RU3 cleaned components (EffNet-B4, s1)	76.7/0.4/22.9	68.8	0.3	72.9	94.9	99 / 92 / 94 / 99				0.5
RU3 + evidence head (EffNet-B4, s1)	83.2/0.6/16.2	62.2	0.5	73.0	94.1	99 / 91 / 93 / 99				0.8
RU4 + part edits (lite) (EffNet-B4, s1)	81.0/0.4/18.6	63.8	0.3	72.6	93.8	99 / 91 / 93 / 99				0.4
RU3 + evidence head, CLIP (CLIP-L/14, s1)	77.4/0.4/22.2	71.4	0.2	74.6	93.8	98 / 90 / 94 / 100				0.2
RU3 + evidence head, CLIP (CLIP-L/14, s2)	73.0/0.3/26.7	73.6	0.2	73.4	95.1	99 / 94 / 96 / 100				0.2
RU4 + part edits, CLIP (main) (CLIP-L/14, s1)	57.9/0.3/41.9	82.3	0.1	70.2	94.8	98 / 94 / 95 / 100				0.0

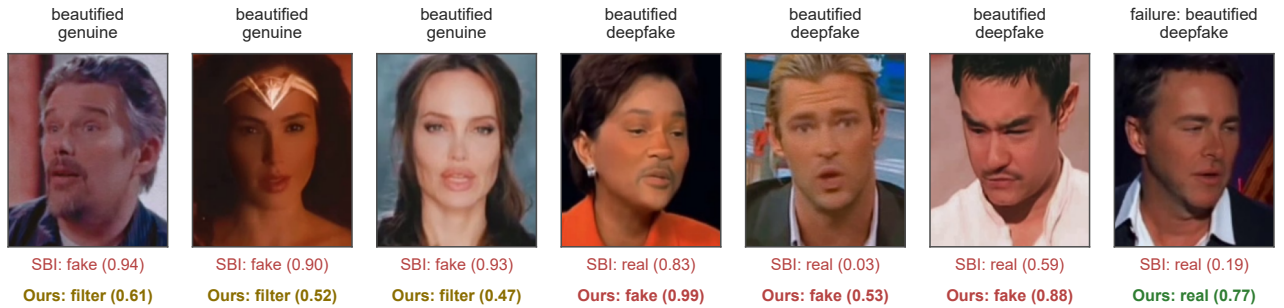


Fig. S8. Celeb-DF-B examples (zero-shot): SBI’s decision at its 5%-FPR threshold on untreated genuine frames ($t = 0.888$, chosen on this set) and our three-way decision. Columns 1–3: beautified genuine faces that SBI accuses and we route to *filter*; 4–6: beautified deepfakes that escape SBI and we call *fake*; 7: a failure case, a deepfake beautified with the *california_dreamin* preset that escapes us too.

TABLE S4

CELEB-DF-B: CHANGE IN DEEPPAKE ESCAPE RATE (PP) PER BEAUTIFICATION PRESET, UNDER COMPRESSION ALONE (SAME RE-ENCODING, NO FILTER), OVERALL, AND OVERALL MINUS THE COMPRESSION EFFECT.

Detector	brown	calif.	hawaii	relax	compress	all	all-compr.
SBI	+43.5	+30.4	+56.0	+11.9	+13.7	+35.5	+21.8
SRM	+19.1	+10.9	+27.8	+10.3	+1.7	+17.0	+15.3
CORE	+9.9	+9.2	+34.7	+1.6	+0.3	+13.8	+13.5
UCF	+19.4	+3.2	+41.1	-14.2	+1.2	+12.4	+11.3
Xception	+7.8	+1.2	+36.0	-5.2	-0.9	+9.9	+10.9
EfficientNet-B4	+11.9	+1.4	+9.0	-9.1	-0.1	+3.4	+3.4
SBI obj., our pipeline	+43.3	+33.5	+45.2	+25.4	+15.3	+36.8	+21.6
Ours HYB s1	+0.3	+10.1	-4.8	+0.1	+0.9	+1.5	+0.6
Ours HYB s2	-1.1	+11.9	-7.5	+1.7	+1.8	+1.3	-0.6
Ours HYB-F	-1.2	+5.2	-3.6	-0.2	-0.3	+0.1	+0.3
Ours HYB-D	-2.7	+5.0	-6.3	-1.9	-0.3	-1.4	-1.2
Ours-S	-7.5	-5.5	-23.0	-11.2	-7.2	-11.8	-4.6

(a) binary SBI (official checkpoint, EffNet-B4)

(b) three-way HYB (ours, EffNet-B4)

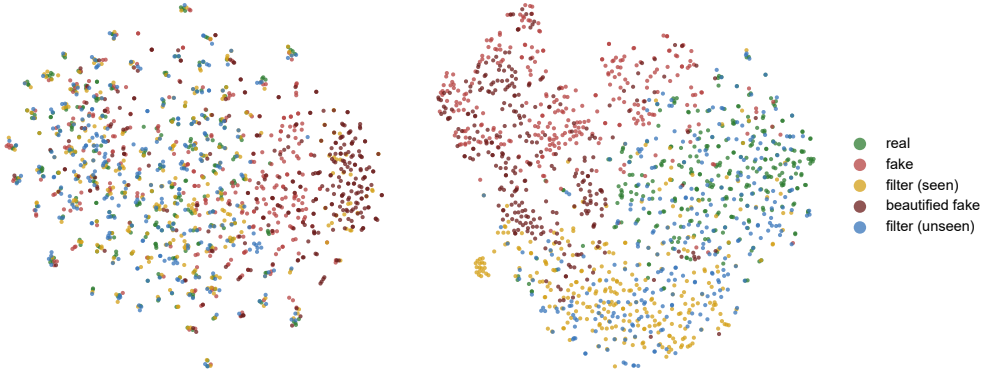


Fig. S9. t-SNE of penultimate features on the same FF++ test images (300 per group). (a) The official SBI checkpoint separates forgeries, but beautified genuine faces are mixed with real faces, as a binary label space requires. (b) HYB forms a filter region, and beautified forgeries stay in the fake region. Of the ten nearest labelled neighbours of a beautified forgery, 11.7% are genuine for SBI and 4.2% for HYB.

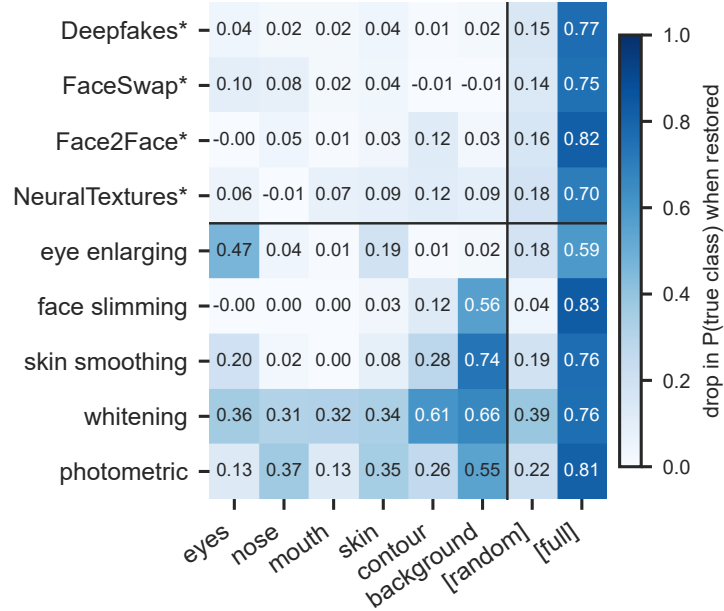


Fig. S10. Region restoration on correctly classified FF++ test images: drop in $P(\text{true class})$ when one facial region is restored to its unedited pixels. [random]: area-matched patch at a random location; [full]: whole image restored. *Forgeries use the real frame of the same video, resized to the fake crop (approximate alignment).

TABLE S5

DETECTION WITH AND WITHOUT THE EVIDENCE HEAD, SAME BACKBONE AND RECIPE; 61.5K PART EDITS IN THE FAKE POOL (THE FIRST CONSISTENCY-LOSS VARIANT SAW 43.4k). CELEB-DF-B: UNTREATED AUC AND % OF UNTREATED GENUINE FRAMES CALLED FAKE.

Variant	CDFv2		DFD		FF++ real/fake/filter	FA / MISS	B1	Celeb-DF-B AUC / accus.
	frame	video	frame	video				
Ours-PE (no head)	0.810	0.871	0.888	0.853	80.7/90.2/76.8	8.9 / 1.6	0/13	0.879 / 56.0
+ evidence head (ours)	0.819	0.878	0.907	0.866	79.0/90.0/84.0	7.0 / 2.1	0/13	0.888 / 60.6
+ evidence head + symmetric degradation	0.817	–	0.896	–	82.8/81.3/79.7	5.5 / 4.8	0/13	0.894 / 17.9
+ consistency loss (43k edits)	0.794	–	0.905	–	83.1/89.1/87.2	3.2 / 2.6	0/13	0.870 / 50.7
+ consistency + symmetric degradation	0.781	–	0.877	–	84.7/80.8/79.6	6.2 / 5.1	0/13	0.875 / 15.1

TABLE S6

FF++ TEST ROUTING OF THE EVIDENCE-HEAD MODEL BY SOURCE (% OF FRAMES SENT TO EACH CLASS; BOLD: CORRECT CLASS).

Class	Source	n	→real	→fake	→filter
real	genuine	1316	79.2	13.8	7.1
fake	Deepfakes	1327	1.0	98.9	0.2
fake	Face2Face	1326	4.0	94.8	1.2
fake	FaceSwap	1323	7.6	92.3	0.2
fake	NeuralTextures	1328	22.6	73.9	3.5
filter	smoothing	156	0.0	7.1	92.9
filter	whitening	149	3.4	8.1	88.6
filter	eye enlarging	152	49.3	13.8	36.8
filter	face slimming	150	16.7	4.7	78.7
filter	photometric (14)	709	2.0	5.8	92.2

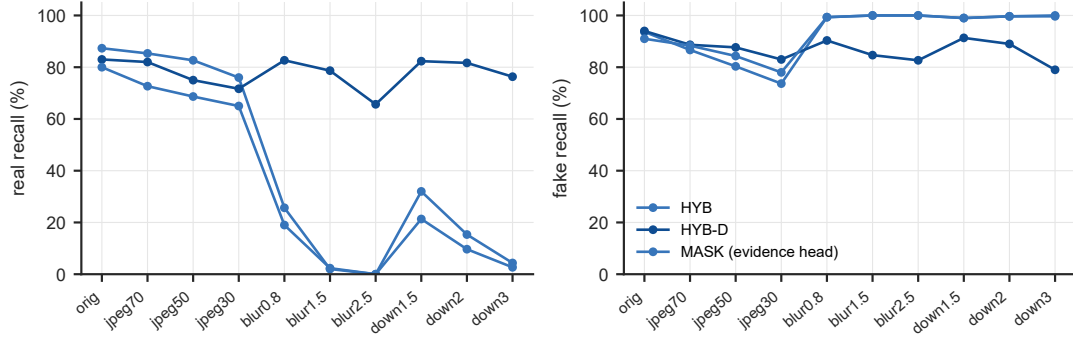


Fig. S11. Recall of genuine (left) and forged (right) FF++ test frames under JPEG re-compression, Gaussian blur and downscaling (300 frames each). Models trained without symmetric degradation call almost every blurred genuine frame fake.

TABLE S7

SINGLE-IMAGE COST (BATCH 1, PyTORCH FP32; CPU: DESKTOP, GPU: RTX 6000 ADA), MEAN OF 50 RUNS.

Model	Params (M)	Input	CPU (ms)	GPU (ms)
RepViT-M0.9 (ours-S)	4.7	224	28.3	8.3
EfficientNet-B4 (HYB)	17.6	380	55.0	10.7
EfficientNet-B4 + evidence head (MASK)	17.6	380	62.1	10.3

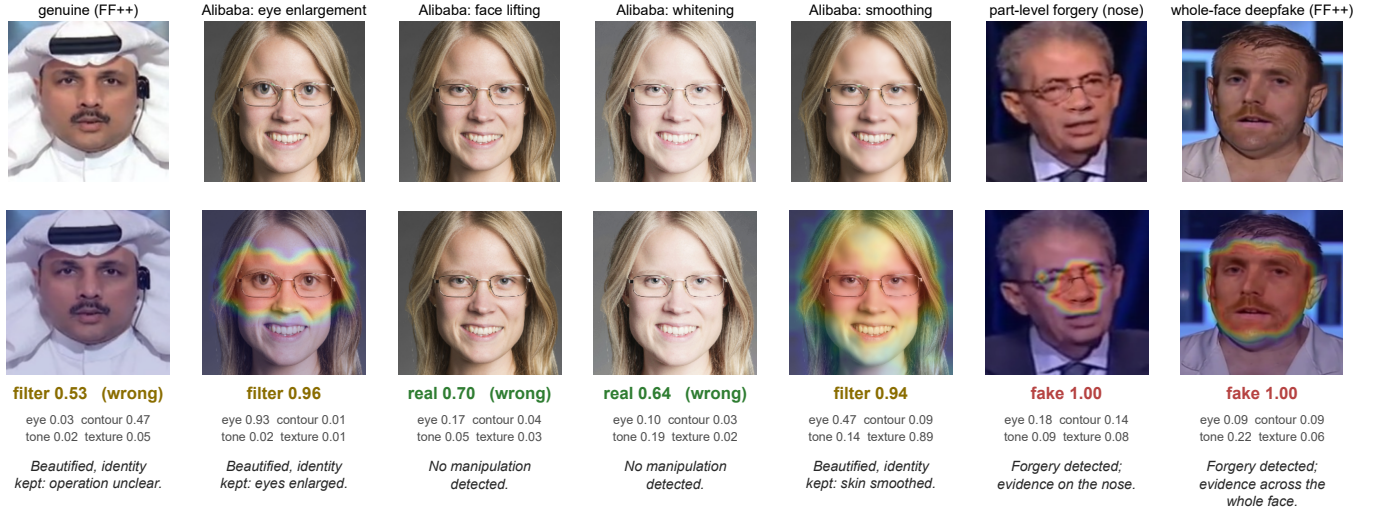


Fig. S12. Output of Ours on examples fixed in advance: one FF++ genuine frame, the four Alibaba renders of one FFHQ face at level 90, a held-out part-level forgery and an FF++ Deepfakes frame. Per image: the three-way decision and its probability, the presence score of each operation, the evidence map, and a template sentence. The sentence names the operations with presence ≥ 0.5 (filter) or the facial part with the highest mean evidence (fake; “whole face” when eyes, nose and mouth all exceed 0.5). No language model is involved. Three of the seven are wrong: lifting and whitening are missed and the genuine frame is sent to *filter*.